

UniVerse

A Campus Companion

Notifications & Push

One insert, two deliveries: a live in-app feed (Supabase Realtime) plus an OS push (Firebase FCM).

Leading University, Sylhet - Dept. of CSE

Course CSE-3240 (Project I) - Team Sherlocked

Flutter (Dart) - Supabase - Firebase FCM - OR-Tools CP-SAT

Owner: Ratul (notifications, push, profile)

Key files: features/notifications/*, core/services/push_service.dart, supabase/functions/send-push

1. The big idea (say this first)

In UniVerse, **inserting one row into the `notifications` table does everything**: it appears live in the in-app feed (Supabase Realtime) AND fires an OS push notification (Firebase FCM) to the right audience. There is exactly one path - no feature sends push directly.

- 1 A source inserts a `notifications` row (admin broadcast, teacher cancel/notice, resource upload, routine publish).
- 2 **In-app**: every signed-in device has a Realtime subscription on the table -> the new row streams in and the feed + unread badge update instantly.
- 3 **Push**: a Supabase DB Webhook on INSERT invokes the `send-push` Edge Function -> it resolves the row's audience, looks up `device\_tokens`, and sends via FCM HTTP v1.

2. Audience targeting

A notification row has `target\_role`, `target\_batch`, `target\_section`. NULLs mean 'everyone'. The client's `matchesAudience()` filters the feed to what applies to this user, and the Edge Function uses the same fields to pick which device tokens to push to. Examples:

targets	audience
<code>target_role=null</code>	Broadcast to the whole campus (e.g. 'routine published').
<code>role=student</code> ,	Only that cohort (a teacher cancelling their class).
<code>role=student</code>	All students (a new resource uploaded).

3. In-app feed details (notification_controller)

- **Read-state** - a separate `notification\_reads` table tracks per-user read marks; the service merges it so each user has their own read/unread state on shared broadcast rows.
- **Unread badge** - `unreadCount` drives the Alerts tab badge in the bottom nav. The controller is app-scoped (one subscription for the whole session).
- **Optimistic mark-read** - tapping marks read in the UI immediately, then writes to the DB, rolling back if it fails.
- **Per-user dismiss** - users can multi-select and hide alerts; dismissed ids are stored locally in SharedPreferences (`dismissed\_notifs\_<uid>`). The shared DB rows are never deleted - dismissal is local only.

4. Push details (push_service.dart)

`PushService` is the only layer touching firebase_messaging + flutter_local_notifications. Why it exists: Realtime can update an OPEN app, but it can't wake a backgrounded or killed app - FCM can.

member	role
<code>init()</code>	Sets up a high-importance Android channel, asks permission, registers handlers.
<code>registerToken(userId)</code>	On sign-in, saves the device's FCM token to `device_tokens` (via a SECURITY DEFINER RPC that clears any stale row first).
<code>foreground messages</code>	Android doesn't auto-show pushes while the app is open, so we draw a local heads-up banner ourselves.
<code>onNotificationTap</code>	Tapping a push routes to the notifications feed.

unregisterToken()

On sign-out, deletes the token BEFORE the session drops (else RLS blocks it).

The channel id ``universe_high_importance`` MUST match AndroidManifest's ``default_notification_channel_id`` or background pushes are silent.

5. The send-push Edge Function (deployed)

A Deno/TypeScript function on Supabase, triggered by the DB Webhook on ``notifications`` INSERT. It resolves the audience -> queries ``device_tokens`` -> sends via **FCM HTTP v1**, minting a service-account JWT in-function (secret ``FCM_SERVICE_ACCOUNT``). The FCM service key lives ONLY here, never in the app.

6. Likely defense questions

question	answer
Realtime vs FCM - why both?	Realtime updates an open app instantly; FCM delivers to backgrounded/killed apps. Together = always delivered.
How does a DB insert send a push?	A Supabase DB Webhook on INSERT invokes the send-push Edge Function automatically.
Where is the FCM secret?	Only inside the Edge Function (<code>FCM_SERVICE_ACCOUNT</code> secret) - never shipped in the APK.
Do dismissed alerts delete data?	No - dismissal is per-user and local (<code>SharedPreferences</code>). Shared broadcast rows are untouched.